

CS 207: Design and Analysis of Algorithms

Practice Questions-2

1. Consider the graph G in Figure 1. You can ignore negative weighted edges for the problems where it is not applicable. Answer the following:
 - (a) Run BFS and DFS on the graph G without considering the weights on the edges. For BFS you can choose to start with the vertex s .
 - (b) Run MST algorithm (both Krushkal's and Prims) and record the attributes of each vertex at the end of each round. What difference can one see running the above two algorithms on weighted and unweighted graphs, in terms of the solution?
 - (c) Run single source shortest paths algorithms from the source vertex s . Record the attributes of each vertex at the end of each round.
 - (d) In each round or iteration, consider the edges in the following order and relax them: $(t, x), (t, y), (t, z), (x, t), (y, x), (y, z), (z, x), (z, s), (s, t), (s, y)$. Correspondingly update the attributes of the vertices in each round in the table below.
 - (e) Run Bellman-Ford's algorithm in the sequence of the above edges and record the attributes at the end of each round.
 - (f) Run APSP algorithms on the graph G and record the corresponding attributes of the vertices at the end of each round.
 - (g) Run Floyd-Warshall's algorithm to find the maximum flow in the network from s to t .
2. Show the correctness of each of the properties used for the Dijkstra's algorithm: triangle inequality, upper-bound property, no-path property, convergence property, and path-relaxation property.
3. Can you give a polynomial time reduction for the following:
 - (a) VERTEX COVER to DOMINATING SET
 - (b) DOMINATING SET to VERTEX COVER
 - (c) VERTEX COVER to INDEPENDENT SET

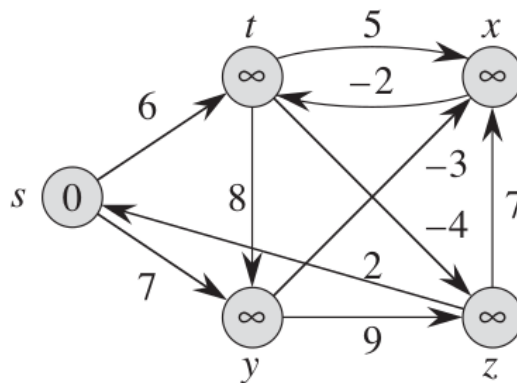


Figure 1: Graph G

- (d) INDEPENDENT SET to VERTEX COVER
- (e) DOMINATING SET to INDEPENDENT SET
- (f) INDEPENDENT SET to DOMINATING SET